25900512

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEMAT201 - GAME THEORY AND PROJECT MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

{Short Answer Questions}

Answer any **5** questions

Each question carries **2** marks

1. What is meant by n-person game? [K] / [CO1]

2. Determine whether the game has a saddle point whose payoff matrix is given by [U] / [CO1]

	Player B	
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$
$\mathbf{A_1}$	1	4
$\mathbf{A_2}$	5	2

3. Find optimal strategies and net outcome using the arithmetic method. [A] / [CO2]

	Player B	
Player A	B_1	B_2
A_1	5	1
A_2	2	4

4. Explain the situations when graphical method is used [U] / [CO2]

5. Draw AON Network for the follwoing list of activities [U] / [CO3]

Activity	*Predecessor Activity*
A	–
B	–
C	–
D	–
E	A, B, C
F	B, C
G	A, C
H	D, E

6. Define cycling in a network [U] / [CO3]

7. How do yo calculate latest occurrence time of the initial event? [K] / [CO4]

8. Define latest allowable time of an event [K] / [CO4]

[2x5 = 10]

Part B

{Short Essay Questions}

Answer any **5** questions

Each question carries **6** marks

9. With an appropriate example, explain maximin and minimax principles. [U] / [CO1]

10. Find the saddle point of the following two-person zero-sum game [U] / [CO1]

	Player B		
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$	$\mathbf{B_3}$
$\mathbf{A_1}$	6	5	7
$\mathbf{A_2}$	5	4	6
$\mathbf{A_3}$	8	5	9

11. Determine the optimal strategies for both players and compute the value of the game for player A using the algebraic method. [A] / [CO2]

	Player B	
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$
$\mathbf{A_1}$	6	1
$\mathbf{A_2}$	-3	2

12. Formulate an equivalent linear programming model for player A and player B based on the zero-sum game defined by the given payoff matrix. [A] / [CO2]

	Player B		
Player A	B_1	B_2	B_3
A_1	6	8	2
A_2	8	2	10
A_3	4	10	12

13. Explain Activity-on-Node network [K] / [CO3]

14. Explain PERT and its significance in Project Management [U] / [CO3]

15. The following table gives the list of various activities and their immediate predecessors involved in a Computer Software Project. [A] / [CO4]

$Activity$	$Predecessor\ activity$	$Duartion(Days)$
A	–	5
B	–	7
C	–	9
D	A, B, C	8
E	A, B, C	5
F	A, B, C	2
G	F	7
H	C	3
I	G	4
J	H, E	3
K	F	5
L	I, J	10

(a) Draw a Network for this project,calculate the critical path.
(b) Identify the critical and noncritical activities

16. Discuss the possible values of Total Float in project scheduling. For each case, explain the reasons for its occurrence and suggest suitable remedies. [U] / [CO4]

[6x5 = 30]

Part C

{Essay Questions}

Answer any **3** questions

Each question carries **10** marks

17. Obtain the strategies for each player and the value of the game for the following two-person zero-sum game by using principle of dominance [U] / [CO1]

	Player B			
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$	$\mathbf{B_3}$	
$\mathbf{A_1}$	8	2	7	6
$\mathbf{A_2}$	3	6	5	6
$\mathbf{A_3}$	5	5	4	5

18. Using the method of row minima and column maxima, solve the following payoff matrix. [E] / [CO2]
Check whether the game has a saddle point. If yes, find the value of the game and strategies of the players:

	Player B		
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$	$\mathbf{B_3}$
$\mathbf{A_1}$	7	3	5
$\mathbf{A_2}$	4	6	2
$\mathbf{A_3}$	8	1	9

19. For the game given by the payoff matrix (Player A): [E] / [CO2]

	$\mathbf{B_1}$	$\mathbf{B_2}$	$\mathbf{B_3}$
$\mathbf{A_1}$	4	-2	3
$\mathbf{A_2}$	2	1	0
$\mathbf{A_3}$	5	2	1

(a) Use the dominance principle to simplify the game.
(b) Solve the reduced game using the matrix method.
(c) Find the value of the game.

20. Following is the list of activities associated with the assembly of a space module for an upcoming mission. [U] / [CO3]
Draw the AOA network diagram of activities involved in the project.

Activity	*Description*	*Predecessor Activity*
A	*Construct shel of module*	–
B	*Order life support systemr*	–
C	*Order components of control*	–
D	*Wire module*	A
E	*Assemble control*	C
F	*Preliminary test*	B
G	*Installmodule*	D, F
H	*Install scientific exp[erimentation*	D, F
I	*Preliminary test of control*	E, F
J	*Instal navigational system*	H, I
K	*Final testing and debugging*	G, J

21. Following table gives the list of various activities and their immediate predecessors involved in the CAT scanner [A] / [CO4]

in a hospital.

$Activity$	$Description$	$Predecessors$	$Duration(days)$
A	$Finalization ofthe layout plans$	$-$	6
B	$Demolition of the structure$	$-$	9
C	$Walls erection$	A, B	15
D	$Flooring$	B	8
E	$Electrical wiring$	C	7
F	$Air conditioning ducting$	C	4
G	$Fire alarm installation$	C	3
H	$False ceiling and light fittings$	E, F, G	10
I	$Wall plastering and painting$	H	9
J	$Equipment installation$	D, I	6
K	$Calibration and testing$	J	3
L	$Final finishing$	K	2
M	$Handing over$	L, K	5

(a) Prepare a network diagram showing the interrelationship of the various activities.
(b) Find the total time required to hold the conference.
(c) Compute the total float for the noncritical activities.

22. The owner of a chain of fast food restaurants is considering a new computer system for accounting and inventory control. A computer company sent the following information about the computer system installation. [A] / [CO4]

$Activity$	$Activity\ Description$	$Predecessors$	$Time\ Duration$
A	$Select\ the\ computer\ model$	$-$	4
B	$Design\ input/output\ system$	A	15
C	$Design\ monitoring\ system$	A	12
D	$Assemble\ computer\ hardware$	B	25
E	$Develop\ themain\ programmes$	B	26
F	$Develop\ input/output\ routines$	C	16
G	$Create\ database$	E	12
H	$Install\ the\ system$	D, F	1
I	$Test\ and\ implement$	G, H	6

(a) Construct the network diagram for the project.
(b) Determine the critical path and compute then expected completion time.
(c) Find Total Float for each activities

[10x3 = 30]